Verigate PortalDot

Authorization middleware for AI-agent payments on PortalDot V3

A proposal is not authority.

Verigate evaluates the action. The bridge verifies signed evidence. A dedicated Solidity guard enforces the exact approved transfer.

VERIGATE	BRIDGE	PORTALDOT
Ed25519 decision	Exact action binding	EIP-712 guard

TESTNET

420420777

PortalDot V3 EVM testnet

V2 transfer recorded • receipt/event matched

Best-fit track

Modular Cloud & Middleware

Built around a narrow, reusable control point between AI intent and token execution.

Preventing unrestricted agent spending

The authorization must bind the decision to the exact thing the executor is allowed to do.

01 RECIPIENT SUBSTITUTION

An approved payment must not be reused with a different destination.

02 AMOUNT TAMPERING

The amount must remain the same from signed intent to contract call.

03 REPLAY OR EXPIRY

A valid authorization should not be usable again or after its deadline.

Design principle

A signed ALLOW is necessary but not enough: its authority must be short-lived, narrowly scoped, and bound to a recipient, amount, chain and contract.

A narrow path from decision to execution

Off-chain verification and on-chain enforcement have separate, explicit trust boundaries.

1

AGENT INTENT

Proposed action

2

VERIGATE

Signed ALLOW + execution authorization

3

BRIDGE

Verify Ed25519; bind exact action

4

EIP-712

Chain, guard, recipient, amount, nonce, deadline

5

GUARD

Reject invalid/replayed or altered transfer

WARN / BLOCK / invalid signature / mismatched action

The bridge refuses to sign; the contract cannot independently override this off-chain decision.

Why PortalDot V3 — and where this project fits

Two 0.1 POT testnet transfer records exist, with different evidence availability for V1 and V2.

- **Recorded testnet transfers (V1 + V2)**

  V1: 0.1 EVM POT in block 132475; saved success receipt and matching TransferExecuted event. A later query to the same configured RPC no longer returned that historical receipt. V2: 0.1 EVM POT in block 149241; receipt, event, sender debit and recipient credit were re-queried through that same RPC.

- **EVM-compatible contract execution**

  The guard uses EIP-712 signatures bound to chain ID and contract address. V2 is the current execution demo; V1 remains a separate predecessor.

- **Substrate-aware operational checks**

  Tooling checks the configured Substrate WebSocket endpoint and genesis hash before sensitive network operations. This is not proof of EVM-to-SS58 account mapping.

BEST-FIT TRACK

Modular Cloud & Middleware

Not claiming:

A cross-chain asset bridge, verified wallet mapping, a second independent EVM RPC provider, production readiness, or an independent audit.

V2 execution evidence on PortalDot V3

The recorded positive path is real; limits and wallet integration are not overclaimed.

NETWORK

PortalDot V3 EVM testnet

CHAIN ID

420420777

CURRENT CONTRACT

PortalDotAgentGuardV2

CONTRACT ADDRESS

0x87C494a5...10E046C6

DEPLOYMENT TX (RECORDED)

0x331c1fc0...3f5d3494b

TRANSFER TX

0xaf3df785...f2e90f5551e

TRANSFER RESULT

Success • block 149241

READ-BACK

V2 policy

Recipient allowlist

0.1 POT / transfer

0.5 POT per resettable 3600-second interval.

TRUST BOUNDARY

Ed25519 stays off-chain; V2 checks the bridge EIP-712 signature.

Live enforcement evidence

Two deployed-contract rejections were observed by read-only eth_call; the successful payment is separate testnet evidence.

UNLISTED RECIPIENT

InvalidRecipient()

The V2 contract rejected an address outside the live allowlist.

OVER-CAP AMOUNT

AmountExceedsCap

0.100000000000000001 POT was rejected against the live 0.1 POT cap.

PUBLIC TRANSFER

SUCCESS • block 149241

A separate 0.1 POT transfer has a matching receipt and TransferExecuted event.

WHAT THE BROWSER VERIFIED

Chain ID 420420777, the recorded V2 runtime hash, authority signer, live policy values, allowlisted demo recipient and the known transfer receipt/event. The denial probes ran at observed block 171216 in the 24 October smoke test.

Verigate ALLOW/BLOCK itself remains a separate local end-to-end integration result.

PROOF BOUNDARY

The two negative results are read-only eth_call responses, not mined reverts. No transaction, signature or fee was involved. One configured RPC was used; the probes do not independently authenticate the operator or prove that Verigate returned BLOCK for those exact probe inputs.

What is proven — and what remains open

Live policy rejections improve the proof; they do not establish production safety or universal wallet control.

- **Sender cost remains unresolved**

  The recipient received 0.1 POT while the sender debit was 7.82782148139542551 POT across the isolated block. It reconciles to gasUsed × effectiveGasPriceWei, but execution fee versus refundable storage deposit is not separated.

- **Independent RPC verification is missing**

  Receipt, runtime hash and denial probes were read through one configured PortalDot RPC. No second-provider check or EVM genesis-hash pin was used for these transfers.

- **Native integration and demand remain unproven**

  A separate native Substrate adapter has offline tests, but no native transfer is claimed. H160-to-SS58 mapping, Wallet/DaaS integration and external user adoption remain unverified.

TRUST AND SCOPE

The guard only controls payments routed through its own balance. It is not a global wallet interceptor.

The admin can change the recipient allowlist and caps. The spending counter resets after its interval; it is not a sliding-window sum. Verigate Ed25519 is verified off-chain by the bridge; the contract verifies the bridge EIP-712 signer.

Testnet prototype • unaudited • not mainnet • no independent RPC authentication.